

Selective Disocclusion Prior Distillation for Feed-Forward Single-View 3D Reconstruction

Anonymous submission

Paper ID ****

Abstract

A generative teacher can improve disoccluded regions, but it is expensive and non-monotonic: on some scenes it is worse than the feed-forward baseline. We study whether its useful behavior can be distilled within a high-teacher-gain regime. We curate training and evaluation scenes using teacher-minus-baseline invisible-region quality measured against ground truth, train an image-space student, and copy baseline pixels in visible regions using masks derived offline from the full target clip. This oracle curation and oracle visibility constitute an upper-bound experiment, not deployable single-view inference. On a 4-scene/161-frame selected holdout, the student improves invisible-region PSNR by +5.77 dB (teacher: +6.09 dB; 0.32 dB gap; 94.7% retained) with exactly zero visible change, while running in **0.087 s/scene versus 247 s**, a $2833\times$ **speedup**. A fixed custom 759-frame diagnostic shows that the student improves LPIPS and FID over the feed-forward evidence component while trailing the teacher on both metrics; it is not an official Flash3D-protocol comparison. A negative ablation shows that recoloring source-plane Gaussians with frozen geometry lacks support at newly disoccluded target pixels. Deployment would require both selection and visibility estimated only from inference-time observables; neither is established here.

1 Introduction

A generative reconstruction teacher guided by feed-forward geometry evidence such as Flash3D [1] can improve disoccluded regions, but requires a slow diffusion denoise. More importantly, the teacher is non-monotonic: it helps hard disocclusions and can hurt easy scenes. The central question is therefore whether useful teacher behavior can be compressed under a clearly delimited high-gain condition.

A naive residual network trained on every teacher render copies both successful and harmful behavior. We instead study *selective disocclusion prior distillation*: scenes are selected using teacher and ground-

truth outcomes, and the student is trained and evaluated within that selected regime. This selection is oracle dataset curation. It does not establish a mechanism for deciding on an unseen scene whether the student should be used.

Contributions.

1. **Conditional principle:** distillation of a non-monotonic teacher on an oracle-curated high-teacher-gain regime, explicitly presented as an upper bound rather than an inference-time selection result.
2. **Model and evidence:** a 46,371-parameter image-space student that achieves +5.77 dB held-out invisible-region gain (teacher: +6.09 dB), exact visible identity, and a $2833\times$ speedup; a fixed custom 759-frame diagnostic reports the quality-speed tradeoff without claiming an official-protocol win.
3. **Architectural diagnosis:** a color-only Gaussian adapter recovers only $\sim 7\%$ of the gap when overfit and gives -1.72 dB on holdout; a support schematic explains why this motivates image-space correction.

2 Method

2.1 Teacher and oracle scene selection

The teacher is the slow selective-injection method: a generative backbone refined by geometry-expert-guided disocclusion injection. It provides high-quality disocclusion targets on hard scenes but is not universally better than the baseline. We compute teacher-minus-baseline invisible-region PSNR against ground truth and retain scenes whose gain exceeds 3 dB. Because this criterion uses the teacher output and ground truth, it is oracle dataset curation and defines an upper-bound condition for training and evaluation.

Table 1: Distillation results (invisible-region PSNR) conditional on oracle high-teacher-gain selection. The feed-forward student retains 94.7% of the teacher’s held-out gain and leaves visible regions exactly unchanged.

Split	Base	Teacher	Student	Stud–Base	Vis Δ
Train (n=696)	15.31	20.68	20.80	+5.49	+0.000
Holdout (n=161)	14.04	20.13	19.81	+5.77	+0.000

2.2 Image-space student

The student operates on rendered frames. Its input is `[f3d_rgb, baseline_rgb, visibility_mask, invisible_mask]`; a small convolutional network predicts a residual r and outputs

$$\text{student} = M \odot \text{baseline} + (1 - M) \odot (\text{f3d} + r),$$

so visible pixels ($M=1$) are copied from the baseline. Training uses invisible-region L1 losses to teacher and GT plus a visible-region identity loss. The visible mask therefore enforces exact no-harm inside each evaluated output; it does not decide whether a scene belongs to the selected regime. These masks are computed offline from the full target clip and are oracle information, not a single-view model prediction.

2.3 Scope of selection

For evaluation on the reported selected holdout, the student is applied directly with the precomputed oracle visibility mask. No learned or rule-based observable selector or online mask estimator is validated here. General use on unseen, uncurated scenes would require both components to use inputs available without teacher renders, target clips, or ground truth.

3 Experiments

All PSNR numbers are **invisible-region** (disocclusion) PSNR, the targeted quantity; visible-region PSNR is a no-harm check.

3.1 Distillation on selected scenes

We train the image-space student on scenes where the teacher beats the baseline by more than 3 dB in invisible-region PSNR, using full per-frame teacher renders, with a 4-scene / 161-frame held-out split drawn from the same oracle-selected regime.

The student retains $5.77/6.09 = 94.7\%$ (approximately 95%) of the teacher’s held-out gain and trails its PSNR by 0.32 dB (19.81 versus 20.13 dB), with visible regions exactly unchanged. The gain is stable across held-out splits (+5.96 and +4.75 dB during scaling).

3.2 Scope of the reported regime

On scenes where the teacher does not outperform the baseline, the student is flat-to-slightly-negative because there is no useful teacher signal to copy. This observation motivates oracle high-gain curation but does not solve selection from observable inputs. Results in this paper must therefore be interpreted as conditional performance on the curated regime.

3.3 Runtime

Table 2: Runtime and quality on the selected regime. The student is four 3×3 convolutions applied once per frame; the teacher requires a 30-step diffusion denoise of the clip.

Method	Time/scene	Inv Δ	Visible
Teacher (30-step diffusion)	247 s	+6.09	lossless
Fast student (CNN)	0.087 s	+5.77	lossless
Speedup	2833$\times$	94.7% kept ($\sim 95\%$)	—

3.4 Fixed custom 759-frame quality–speed diagnostic

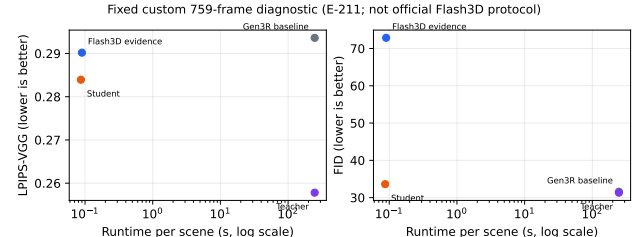


Figure 2: Quality–speed vectors from the released E-211 JSON on the same fixed custom 21-scene/759-frame disocclusion-heavy diagnostic (256 $\times$ 256, VGG-LPIPS). Flash3D is the geometry/evidence component measured within this custom diagnostic, not an official-protocol benchmark opponent. Lower is better on both vertical axes. The student improves over the evidence component at feed-forward cost but trails the teacher on LPIPS and FID.

Table 3: Exact rounded values behind Fig. 2. These same-diagnostic measurements support component analysis only; they do not establish an official-protocol ranking.

Method	PSNR $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	Time
Gen3R baseline	17.88	.294	31.6	247 s
Flash3D evidence	19.98	.290	72.9	0.09 s
Teacher (slow)	19.37	.258	31.3	247 s
Student (ours)	18.17	.284	33.6	0.087 s

Within this fixed diagnostic, the student records lower LPIPS (0.284 versus 0.290) and FID (33.6 versus 72.9) than the Flash3D evidence component at similar measured runtime, but trails the teacher (LPIPS 0.258, FID 31.3). We deliberately restrict this statement to E-211: protocol differences preclude interpreting it as an official Flash3D result. The released JSON and `scripts/generate_quality_speed.py` reproduce the quality-speed scatter.

Released checkpoint. The package includes the student state dictionary, a model card documenting its intended use and limitations, and a validation script that verifies the SHA256, 46,371-parameter architecture, state dictionary, and synthetic output shape.

3.5 Ablation: why not intervene at the Gaussian layer?

A natural alternative edits the feed-forward backbone’s Gaussian attributes directly. E-150 tests a color-only adapter that changes source-plane `features_dc` and re-renders while freezing means, opacity, scale, and rotation. It reaches only $\sim 7\%$ of the teacher-baseline gap when overfit on one scene and does not generalize (holdout -1.72 dB). The lower portion of Fig. 1 states exactly what this evidence supports: recoloring existing source-plane Gaussians does not add geometric support where a target view newly reveals content. The schematic is not a qualitative render result and does not claim that all Gaussian-space adapters fail; it explains why this tested color-only, frozen-geometry intervention has limited leverage. We therefore use an image-space residual over the geometry evidence.

4 Discussion

Within the oracle-selected regime, a tiny feed-forward student retains 94.7% (approximately 95%) of the teacher’s disocclusion gain at $\sim 3000\times$ lower cost with exact visible-region identity. The fixed custom diagnostic characterizes this quality-speed tradeoff, while the Gaussian-color ablation explains the chosen intervention level. The result establishes conditional compressibility of useful teacher behavior, not automatic identification of where that behavior is useful.

5 Limitations

Final numbers use a high-teacher-gain scene set selected with teacher and ground-truth outcomes and visibility masks computed from the full target clip.

Both oracle inputs are unavailable for ordinary single-view inference and may overstate end-to-end performance relative to a practical system. The reported run used the Flash3D disocclusion basis, offline frame-label weighting, and a 4-scene/161-frame holdout. Released checkpoint evaluation is reproducible, but exact retraining is partial rather than a self-contained command: external teacher arrays, the frame-label JSON, and the original holdout manifest are not included, and the exact scene IDs and frame-label path are unknown in this release. The released training manifest records the known settings and a placeholder-only command template. No observable inference-time selector or online visibility estimator is established. Extending the method beyond the selected regime requires learning and validating both. An optional stronger adapter could predict residual disocclusion Gaussians (new points) rather than image-space residuals; this is separate future work.

6 Conclusion

On an oracle-curated high-teacher-gain regime, the student recovers $+5.77$ dB of invisible-region PSNR (teacher: $+6.09$ dB; 0.32 dB gap; 94.7% retained) while running $\sim 3000\times$ faster. The visible mask leaves baseline pixels exactly unchanged inside the evaluated output. The fixed custom 759-frame diagnostic and Gaussian color-adapter negative ablation delimit what the evidence supports. Generalization to uncurated scenes remains contingent on future observable selection and visibility estimation.

References

- [1] Stanislaw Szymanowicz, Eldar Insafutdinov, Chuanxia Zheng, Dylan Campbell, Joao Henriques, Christian Rupprecht, and Andrea Vedaldi. Flash3d: Feed-forward generalisable 3d scene reconstruction from a single image. In *arXiv preprint arXiv:2406.04343*, 2024.

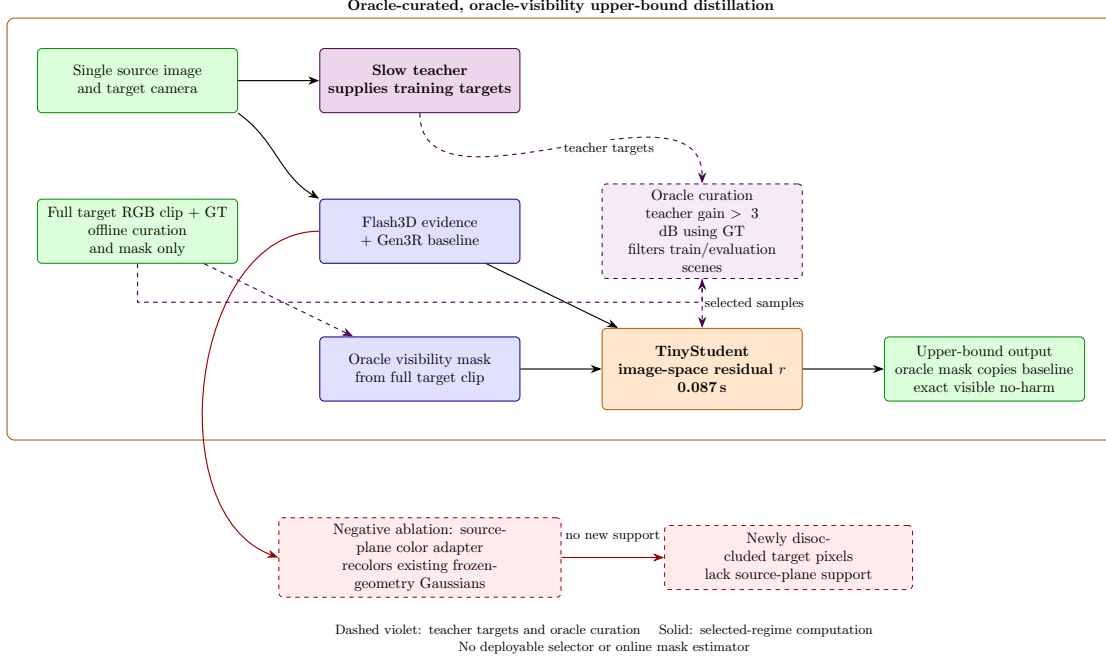


Figure 1: Oracle-selected distillation of a non-monotonic teacher. Teacher and ground-truth outcomes select high-gain scenes for training and evaluation, while the full target clip supplies oracle visibility. Within this selected regime, the 46,371-parameter CNN predicts an image-space residual from geometry evidence, the baseline, and visibility; the visible mask copies baseline pixels exactly. There is no inference-time selection or online mask-estimation mechanism in the reported pipeline. The lower schematic summarizes the E-150 negative ablation: a source-plane color adapter can recolor existing frozen-geometry Gaussians but cannot create support for newly disoccluded target pixels. It is an explanation of the measured ablation, not an invented render comparison.